



# Odour emissions control and dispersion analysis of $H_2S$ concentrations in a WWTP

## CLIENT

Shafdan WWTP

## SECTOR

Industry

## LOCATION

Rishon LeZion, Israel

## PRODUCTS

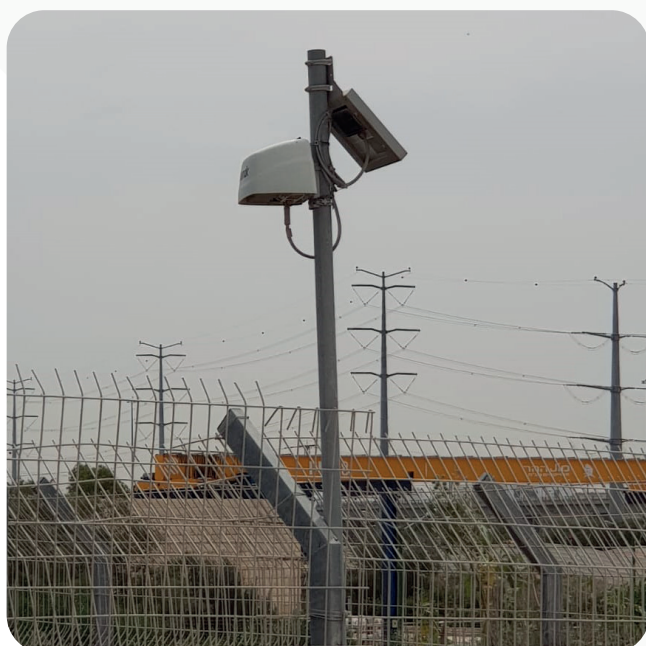
Kunak AIR Pro  
Kunak AIR Lite  
Kunak AIR Cloud

## Need

Rishon LeZion municipality claims that the operation of the Wastewater Treatment Plant (WWTP) affects the residential areas around the plant with odours due to the emission of  $H_2S$ .

The client has invested a lot of resources in order to prevent such emissions and wants to ensure that  $H_2S$  levels at the perimeter of the plant are according to the Clean Air Act or to get early alerts in case of any deviation.

Meteo-Tech, the official distributor of Kunak solutions for Israel, thanks to its extensive experience in environmental and meteorological monitoring, was in charge of deploying a fence line monitoring network to control  $H_2S$  emissions derived from the wastewater treatment plant activities that could affect nearby populations.





When H<sub>2</sub>S episodes occur, Kunak AIR sensors can detect them and Kunak AIR Cloud sends alerts when thresholds are exceeded.



## Solution

Deployment of 7 Kunak AIR stations on the perimeter of the Shafdan Wastewater Treatment Plant (fence-line monitoring nearby the residential areas).

Additionally, an H<sub>2</sub>S analyzer located in a short chimney will be able to compute, according to the Meteo-Tech patent, the plume dispersion and which areas are above the smell threshold and may cause odour nuisance in some neighbourhoods and thus verify or negate complaints.

## Results

The client has continuous measurements of H<sub>2</sub>S levels at the perimeter of the plant and knows the situation with H<sub>2</sub>S emissions with a high level of confidence and validity.

The Kunak sensors have been installed in front of the neighbourhoods where odour complaints come from. When the H<sub>2</sub>S episodes occur and are detected through Kunak AIR Cloud thresholds, the Meteo-tech's patent plume computes the emission rate to be able to check the in situ chimney concentration and to determine that the H<sub>2</sub>S came either from the WWTP or another source in the area.

## Benefits

- Accurate and continuous H<sub>2</sub>S monitoring.
- Detection of odour episodes.
- Control of emissions produced by wastewater processing.
- Possibility to deny complaints with reliable data.
- Real-time information on concentrations for decision-making.
- Evaluation of the efficiency of the control systems in place.

